

Informe de experimento: results_probe_forced

Generado el 2026-09-15 22:05 por benchmarks/experiment_report.py a partir de 2 report(s) del directorio results_probe_forced (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f  
↪ low-agent/benchmarks/results_probe_forced
```

- Reports leídos: 2 (report.rescored.json usado en 0; el resto report.json)
- Modelos: openrouter:openai/gpt-4o, openrouter:openai/gpt-4o-mini
- Métodos: llm_only_forced:cot, llm_only_forced:structured
- Casos: case14, case30
- Requests: generadas, N=20 por caso y seed, difficulties=todas (2 reports)
- Items por dificultad: plain=40, parameterized=40, multistep=40, ambiguous=40
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-15 20:54 –2026-09-15 20:58
- Costo total: 2.0979 USD (160 items; costo medio 0.01311 USD/item)
- Items totales: 160; ok=152; con error=8; tokens totales=771643
- Solved global: 0.6 % (1/160)

Errores por tipo:

Tipo	Items	Métodos afectados
json_parse_failed	8	llm_only_forced:cot (5), llm_only_forced:structured (3)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

Método llm_only_forced:cot

System prompt —860 caracteres; strategy=cot, forced=True, sin herramientas

You are a power system analysis expert. You need to calculate AC power flow results based on test system data provided by the user.

Important Constraints:

- You must strictly output JSON (no extra text, no Markdown blocks).
- Do not output NaN/Infinity; all values must be finite real numbers.
- You must compute and report your best numerical estimate for every bus voltage and every branch flow. `converged` must be true and every array must be complete (one entry per bus and per branch). Do not leave arrays empty and do not state that you cannot compute.

Reasoning mode: you MAY write a short step-by-step reasoning in plain text before the answer. The final JSON object must be the LAST thing in the response, inside a single ```json fenced block, and it must be the only JSON object in the response. Do not use curly braces in the reasoning text.

User prompt (primer item) —6461 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

You are a power systems expert. Based on the following IEEE case14 test system data, calculate the AC power flow results.

System Data (from pandapower network tables)

Node Data (Bus Table: net.bus)

	name	vn_kv	type	zone	in_service
0	1	135.000	b	1.0	True
1	2	135.000	b	1.0	True
2	3	135.000	b	1.0	True
3	4	135.000	b	1.0	True
4	5	135.000	b	1.0	True
5	6	0.208	b	1.0	True

```

6   7   14.000   b   1.0       True
7   8   12.000   b   1.0       True
8   9    0.208   b   1.0       True
9  10    0.208   b   1.0       True
10 11    0.208   b   1.0       True
... (3 líneas omitidas de 15)

```

Load Data (Load Table: net.load)

	bus	p_mw	q_mvar	in_service
0	1	21.473798	12.567615	True
1	2	85.726702	17.290948	True
2	3	45.568126	-3.717902	True
3	4	6.887322	1.449962	True
4	5	11.740497	7.861940	True
5	8	32.229269	18.135792	True
6	9	9.714580	6.260507	True
7	10	3.424253	1.761044	True
8	11	6.539657	1.715320	True
9	12	14.757610	6.340306	True
10	13	15.783969	5.296634	True

Generator Data (Generator Table: net.gen)

	bus	p_mw	vm_pu	min_p_mw	max_p_mw	min_q_mvar	max_q_mvar	in_service
0	1	41.63323	1.045	0.0	140.0	-40.0	50.0	True
1	2	0.00000	1.010	0.0	100.0	0.0	40.0	True
2	5	0.00000	1.070	0.0	100.0	-6.0	24.0	True
3	7	0.00000	1.090	0.0	100.0	-6.0	24.0	True

Balanced Node/External Grid (External Grid Table: net.ext_grid)

	bus	vm_pu	va_degree	in_service
0	0	1.06	0.0	True

Line Data (Line Table: net.line)

	from_bus	to_bus	length_km	r_ohm_per_km	x_ohm_per_km	c_nf_per_km	max_i_ka	df	parallel	in_service
0	0	1	1.0	3.532005	10.783732	768.484773	42.339020	1.0	1	True
1	0	4	1.0	9.846967	40.649040	716.088084	42.339020	1.0	1	True
2	1	2	1.0	8.563928	36.080033	637.493051	42.339020	1.0	1	True
3	1	3	1.0	10.590547	32.134320	494.857619	42.339020	1.0	1	True
4	1	4	1.0	10.379138	31.689630	503.590401	42.339020	1.0	1	True
5	2	3	1.0	12.212573	31.170217	186.299339	42.339020	1.0	1	True
6	3	4	1.0	2.433038	7.674548	0.000000	42.339020	1.0	1	True
7	5	10	1.0	0.000041	0.000086	0.000000	27479.652235	1.0	1	True
8	5	11	1.0	0.000053	0.000111	0.000000	27479.652235	1.0	1	True
9	5	12	1.0	0.000029	0.000056	0.000000	27479.652235	1.0	1	True
10	8	9	1.0	0.000014	0.000037	0.000000	27479.652235	1.0	1	True

... (4 líneas omitidas de 16)

Transformer Data (Transformer Table: net.trafo)

	hv_bus	lv_bus	sn_mva	vn_hv_kv	vn_lv_kv	vk_percent	vk_r_percent	pfe_kw	i0_percent	shift_degree	tap_side	tap_neutral	tap_min
↪	tap_max	tap_step_percent	in_service										
0	3	6	9900.0	135.0	14.000	2070.288	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN	2.2	True										
1	3	8	9900.0	135.0	0.208	5506.182	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN	3.1	True										
2	4	5	9900.0	135.0	0.208	2494.998	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN	6.8	True										
3	6	7	9900.0	14.0	12.000	1743.885	0.0	0.0	0.0	0.0	None	NaN	NaN
↪	NaN	NaN	True										
4	6	8	9900.0	14.0	0.208	1089.099	0.0	0.0	0.0	0.0	None	NaN	NaN
↪	NaN	NaN	True										

Task

Load case14 and disconnect the line between bus ten and bus eleven.

Reasoning Instructions

Before producing the final JSON, reason step by step:

1. Which buses does the request refer to (display ids, slack bus, load and generator buses)?
 2. Which lines/branches are involved (from_bus-to_bus, line_id convention)?
 3. Which quantities and units are asked for (MW, Mvar, p.u., degrees, % loading)?
 4. What changes with respect to the base case, and which power balance must hold?
- Write the reasoning first, then the final JSON as the last part of the response.

Output Requirements

Please calculate and return the following results (JSON format):

- 1) Voltage magnitude (p.u.) and phase angle (degrees) for each node
 - 2) Active power (MW) and load factor (%) at the beginning of each branch
- For lines: `line\_id` uses the row index of `net.line`
 - For transformers: `line\_id` uses the row index of 100000 + `net.trafo`

3) Total generation (MW), total load (MW), total losses (MW)

4) Are there any voltage overruns (<0.95 or >1.05 p.u.) or line overruns (>100%)?

Please strictly follow the following JSON schema output (all fields are complete and of correct type):

```
{
  "converged": true/false,

  "bus_voltages": [{"bus_id": 1, "vm_pu": 1.0, "va_deg": 0.0}, ...],

  "line_flows": [{"line_id": 0, "p_from_mw": 0.0, "loading_percent": 0.0}, ...],

  "total_generation_mw": 0.0,

  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
```

[Important] bus_id must use the `name` column of net.bus (the usual IEEE/MATPOWER 1..N numbering), not the row index of net.bus.

Método llm_only_forced:structured

System prompt —565 caracteres; strategy=structured, forced=True, sin herramientas

You are a power system analysis expert. You need to calculate AC power flow results based on test system data provided by the user.

Important Constraints:

- You must strictly output JSON (no extra text, no Markdown blocks).
- Do not output NaN/Infinity; all values must be finite real numbers.
- You must compute and report your best numerical estimate for every bus voltage and every branch flow. `converged` must be true and every array must be complete (one entry per bus and per branch). Do not leave arrays empty and do not state that you cannot compute.

User prompt (primer item) —5966 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

You are a power systems expert. Based on the following IEEE case14 test system data, calculate the AC power flow results.

System Data (from pandapower network tables)

Node Data (Bus Table: net.bus)

	name	vn_kv	type	zone	in_service
0	1	135.000	b	1.0	True
1	2	135.000	b	1.0	True
2	3	135.000	b	1.0	True
3	4	135.000	b	1.0	True
4	5	135.000	b	1.0	True
5	6	0.208	b	1.0	True
6	7	14.000	b	1.0	True
7	8	12.000	b	1.0	True
8	9	0.208	b	1.0	True
9	10	0.208	b	1.0	True
10	11	0.208	b	1.0	True
...	(3 líneas omitidas de 15)				

Load Data (Load Table: net.load)

	bus	p_mw	q_mvar	in_service
0	1	21.473798	12.567615	True
1	2	85.726702	17.290948	True
2	3	45.568126	-3.717902	True
3	4	6.887322	1.449962	True
4	5	11.740497	7.861940	True
5	8	32.229269	18.135792	True
6	9	9.714580	6.260507	True
7	10	3.424253	1.761044	True
8	11	6.539657	1.715320	True
9	12	14.757610	6.340306	True
10	13	15.783969	5.296634	True

Generator Data (Generator Table: net.gen)

	bus	p_mw	vm_pu	min_p_mw	max_p_mw	min_q_mvar	max_q_mvar	in_service
0	1	41.63323	1.045	0.0	140.0	-40.0	50.0	True
1	2	0.00000	1.010	0.0	100.0	0.0	40.0	True
2	5	0.00000	1.070	0.0	100.0	-6.0	24.0	True
3	7	0.00000	1.090	0.0	100.0	-6.0	24.0	True

Balanced Node/External Grid (External Grid Table: net.ext_grid)

	bus	vm_pu	va_degree	in_service
--	-----	-------	-----------	------------

```
0 0 1.06 0.0 True
```

```
### Line Data (Line Table: net.line)
```

	from_bus	to_bus	length_km	r_ohm_per_km	x_ohm_per_km	c_nf_per_km	max_i_ka	df	parallel	in_service
0	0	1	1.0	3.532005	10.783732	768.484773	42.339020	1.0	1	True
1	0	4	1.0	9.846967	40.649040	716.088084	42.339020	1.0	1	True
2	1	2	1.0	8.563928	36.080033	637.493051	42.339020	1.0	1	True
3	1	3	1.0	10.590547	32.134320	494.857619	42.339020	1.0	1	True
4	1	4	1.0	10.379138	31.689630	503.590401	42.339020	1.0	1	True
5	2	3	1.0	12.212573	31.170217	186.299339	42.339020	1.0	1	True
6	3	4	1.0	2.433038	7.674548	0.000000	42.339020	1.0	1	True
7	5	10	1.0	0.000041	0.000086	0.000000	27479.652235	1.0	1	True
8	5	11	1.0	0.000053	0.000111	0.000000	27479.652235	1.0	1	True
9	5	12	1.0	0.000029	0.000056	0.000000	27479.652235	1.0	1	True
10	8	9	1.0	0.000014	0.000037	0.000000	27479.652235	1.0	1	True

```
... (4 líneas omitidas de 16)
```

```
### Transformer Data (Transformer Table: net.trafo)
```

	hv_bus	lv_bus	sn_mva	vn_hv_kv	vn_lv_kv	vk_percent	vk_r_percent	pfe_kw	i0_percent	shift_degree	tap_side	tap_neutral	tap_min
0	↪ tap_max	tap_step_percent	in_service										
↪	3	6	9900.0	135.0	14.000	2070.288	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN	2.2	True										
1	3	8	9900.0	135.0	0.208	5506.182	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN	3.1	True										
2	4	5	9900.0	135.0	0.208	2494.998	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN	6.8	True										
3	6	7	9900.0	14.0	12.000	1743.885	0.0	0.0	0.0	0.0	None	NaN	NaN
↪	NaN	NaN	True										
4	6	8	9900.0	14.0	0.208	1089.099	0.0	0.0	0.0	0.0	None	NaN	NaN
↪	NaN	NaN	True										

```
## Task
```

```
Load case14 and disconnect the line between bus ten and bus eleven.
```

```
## Output Requirements
```

```
Please calculate and return the following results (JSON format):
```

```
1) Voltage magnitude (p.u.) and phase angle (degrees) for each node
```

```
2) Active power (MW) and load factor (%) at the beginning of each branch
```

```
- For lines: `line_id` uses the row index of `net.line`
```

```
- For transformers: `line_id` uses the row index of 100000 + `net.trafo`
```

```
3) Total generation (MW), total load (MW), total losses (MW)
```

```
4) Are there any voltage overruns (<0.95 or >1.05 p.u.) or line overruns (>100%)?
```

```
Please strictly follow the following JSON schema output (all fields are complete and of correct type):
```

```
{
  "converged": true/false,

  "bus_voltages": [{"bus_id": 1, "vm_pu": 1.0, "va_deg": 0.0}, ...],

  "line_flows": [{"line_id": 0, "p_from_mw": 0.0, "loading_percent": 0.0}, ...],

  "total_generation_mw": 0.0,

  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
```

```
[Important] bus_id must use the `name` column of net.bus (the usual IEEE/MATPOWER 1..N numbering), not the row index of net.bus.
```

3. Resultados agregados

Columnas del protocolo (medias por método × caso, del `scoreboard_per_case`): Solved = respondió correctamente de extremo a extremo; Form. = formulación exacta de las tool calls; V_MAE (p.u.) y F_MAE (MW) sobre los items con resultado; FR = tasa de convergencia/feasibility; Faith. = números de la respuesta trazables a salidas de tools; SFR = fallos declarados con seguridad; Claim = éxito reclamado sobre un fallo; Abst. = abstención en items resolubles; Calls = tool calls medias; Tok. = tokens medios; TTV = tiempo medio (s); \$ = costo total USD. Porcentajes en %.

Por método × caso

Modelos: M1 = openrouter:openai/gpt-4o; M2 = openrouter:openai/gpt-4o-mini

Model	Method	Case	N	Solved	Form.	V_MAE	F_MAER	Faith.	SFR	Claim	Abst.	Calls	Tok.	TTV	\$	
M1	llm_only_forced:cot	case14	20	5.0	–	2.55e-02	32.02	100.0	4.6	–	–	0.0	n/a	3959	10.4	0.3987
M1	llm_only_forced:cot	case30	20	0.0	–	1.38e-02	16.08	100.0	4.6	–	–	0.0	n/a	6280	15.7	0.6671
M1	llm_only_forced:structured	case14	20	0.0	–	3.56e-02	29.35	100.0	3.5	–	–	0.0	n/a	3457	7.0	0.3247
M1	llm_only_forced:structured	case30	20	0.0	–	1.42e-02	22.26	100.0	5.6	–	–	0.0	n/a	5776	13.0	0.5929
M2	llm_only_forced:cot	case14	20	0.0	–	3.41e-01	32.40	100.0	4.6	–	–	5.0	n/a	3850	11.3	0.0226
M2	llm_only_forced:cot	case30	20	0.0	–	2.32e-02	20.46	100.0	12.5	–	–	0.0	n/a	6178	19.3	0.0388
M2	llm_only_forced:structured	case14	20	0.0	–	4.54e-01	31.87	100.0	1.3	–	–	0.0	n/a	3325	8.7	0.0179
M2	llm_only_forced:structured	case30	20	0.0	–	2.04e-02	10.58	100.0	13.2	–	–	0.0	n/a	5758	19.8	0.0354

Agregado por método (media ponderada por items sobre todos los casos)

Modelos: M1 = openrouter:openai/gpt-4o; M2 = openrouter:openai/gpt-4o-mini

Model	Method	Case	N	Solved	Form.	V_MAE	F_MAER	Faith.	SFR	Claim	Abst.	Calls	Tok.	TTV	\$	
M1	llm_only_forced:cot	2 casos	40	2.5	–	1.96e-02	24.05	100.0	4.6	–	–	0.0	n/a	5119	13.0	1.0657
M1	llm_only_forced:structured	2 casos	40	0.0	–	2.49e-02	25.80	100.0	4.6	–	–	0.0	n/a	4617	10.0	0.9175
M2	llm_only_forced:cot	2 casos	40	0.0	–	1.59e-01	25.58	100.0	8.5	–	–	2.5	n/a	5014	15.3	0.0614
M2	llm_only_forced:structured	2 casos	40	0.0	–	2.20e-01	20.36	100.0	7.2	–	–	0.0	n/a	4541	14.3	0.0532

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	plain	parameterized	multistep	ambiguous
llm_only_forced:cot	n/a	n/a	n/a	n/a
llm_only_forced:structured	n/a	n/a	n/a	n/a

Solved por dificultad (% , items resueltos / items)

Method	plain	parameterized	multistep	ambiguous
llm_only_forced:cot	5.0 (1/20)	0.0 (0/20)	0.0 (0/20)	0.0 (0/20)
llm_only_forced:structured	0.0 (0/20)	0.0 (0/20)	0.0 (0/20)	0.0 (0/20)

Tipos de error de formulación (conteo de items)

Method	unparsed
llm_only_forced:cot	5
llm_only_forced:structured	3

Errores de ejecución por método (conteo de items con error)

Method	json_parse_failed
llm_only_forced:cot	5
llm_only_forced:structured	3

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed_success_on_failure, stale_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de **traces/** cuando existe.

Método llm_only_forced:cot

Ejemplo: resuelto (solved=True)

```
request (case14-plain-012-s0, case14, difficulty=plain, seed=1683194653): Load the 14-bus test case (case14).
intended_calls: load_case({"case_name": "case14"})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
To solve the AC power flow for the IEEE 14-bus test system, we need to follow these steps:
```

```

1. **Identify Buses:**
  - Slack Bus: Bus 0 (External Grid)
  - Load Buses: Buses 1, 2, 3, 4, 5, 8, 9, 10, 11, 12, 13
  - Generator Buses: Buses 1, 2, 5, 7

2. **Identify Lines/Branches:**
  - Lines: From the `net.line` table, lines are identified by their `from_bus` and `to_bus` columns.
  - Transformers: From the `net.trafo` table, transformers are identified by their `hv_bus` and `lv_bus` columns.

3. **Quantities and Units:**
  - Voltage Magnitude (p.u.) and Phase Angle (degrees) for each bus.
  - Active Power (MW) and Load Factor (%) at the beginning of each branch.

4. **Power Balance:**
  - Total generation must equal total load plus losses.
  - Ensure no voltage or line loading limi ...[2535 chars más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=8.49e-04; F_MAE=24.26; faithful=0.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=3944;
↪ cost_usd=0.0199; wall=8.6 s

```

Ejemplo: con bandera de fallo (claimed/stale/abstained)

```

request (case14-ambiguous-007-s0, case14, difficulty=ambiguous, seed=2073320062): Load the 14-bus test case (case14) and disconnect the
↪ line between bus 6 and bus 13 (1-based).
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
1. The request refers to the following buses:
  - Slack bus: Bus 1 (external grid)
  - Load buses: Buses 1, 2, 3, 4, 5, 8, 9, 10, 11, 12, 13
  - Generator buses: Buses 1, 2, 5, 7

2. The line being disconnected is between bus 6 and bus 13, which corresponds to line_id 12 in the line data.

3. The quantities requested include:
  - Voltage magnitude (p.u.) and phase angle (degrees) for each bus
  - Active power (MW) and load factor (%) at the beginning of each branch
  - Total generation (MW), total load (MW), and total losses (MW)

4. With the disconnection of the line between bus 6 and bus 13, bus 13 will be isolated from the network. This will affect the power
↪ balance, as the loads at bus 13 will not be supplied. The total generation will need to be recalculated without the contribu ...[2096
↪ chars más]
--- Scoring ---
formulation=unparsed (exact=None); V_MAE=--; F_MAE=--; faithful=0.0 %; solved=False (run_failed); flags=abstained,abstained_on_solvable;
↪ error=json_parse_failed; tokens=3805; cost_usd=0.0011; wall=12.0 s

```

Ejemplo: no resuelto

```

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↪ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
1. The request refers to the following buses:
  - Slack bus: Bus 0 (external grid)
  - Load buses: Buses 1, 2, 3, 4, 5, 8, 9, 10, 11, 12, 13
  - Generator buses: Buses 1, 2, 5, 7

2. The line involved in the request is the one between bus 10 and bus 11, which is line_id 13.

3. The quantities and units requested are:
  - Voltage magnitude (p.u.) and phase angle (degrees) for each bus
  - Active power (MW) and load factor (%) at the beginning of each branch
  - Total generation (MW), total load (MW), and total losses (MW)

4. With the disconnection of the line between bus 10 and bus 11, the power flow will change, affecting the load distribution and
↪ potentially leading to changes in voltage levels and branch flows. The power balance must hold, meaning total generation must equal
↪ total ...[1843 chars más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=4.38e-01; F_MAE=37.44; faithful=0.0 %; solved=False (voltage_error); flags=-; error=-; tokens=3687;
↪ cost_usd=0.0010; wall=12.3 s

```

Método llm_only_forced:structured

Ejemplo: no resuelto

```

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↪ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)

```

```

--- Respuesta final ---
{
  "converged": true,
  "bus_voltages": [
    {"bus_id": 1, "vm_pu": 1.045, "va_deg": 0.0},
    {"bus_id": 2, "vm_pu": 1.010, "va_deg": -2.0},
    {"bus_id": 3, "vm_pu": 1.020, "va_deg": -5.0},
    {"bus_id": 4, "vm_pu": 1.015, "va_deg": -3.0},
    {"bus_id": 5, "vm_pu": 1.070, "va_deg": 0.0},
    {"bus_id": 6, "vm_pu": 0.208, "va_deg": 0.0},
    {"bus_id": 7, "vm_pu": 1.090, "va_deg": 0.0},
    {"bus_id": 8, "vm_pu": 1.020, "va_deg": -1.0},
    {"bus_id": 9, "vm_pu": 0.208, "va_deg": 0.0},
    {"bus_id": 10, "vm_pu": 0.208, "va_deg": 0.0},
    {"bus_id": 11, "vm_pu": 0.208, "va_deg": 0.0},
    {"bus_id": 12, "vm_pu": 0.208, "va_deg": 0.0},
    {"bus_id": 13, "vm_pu": 0.208, "va_deg": 0.0},
    {"bus_id": 14, "vm_pu": 0.208, "va_deg": 0.0}
  ],
  "line_flows": [
    {"line_id": 0, "p_from ...[997 chars más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=4.36e-01; F_MAE=35.45; faithful=0.0 %; solved=False (voltage_error); flags=-; error=-; tokens=3293;
↪ cost_usd=0.0009; wall=10.1 s

```

Ejemplo: primer item

```

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↪ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
```json
{
 "converged": true,
 "bus_voltages": [
 {"bus_id": 1, "vm_pu": 1.045, "va_deg": 0.0},
 {"bus_id": 2, "vm_pu": 1.010, "va_deg": -4.98},
 {"bus_id": 3, "vm_pu": 1.070, "va_deg": -12.72},
 {"bus_id": 4, "vm_pu": 1.090, "va_deg": -10.33},
 {"bus_id": 5, "vm_pu": 1.070, "va_deg": -8.78},
 {"bus_id": 6, "vm_pu": 1.020, "va_deg": -14.22},
 {"bus_id": 7, "vm_pu": 1.090, "va_deg": -13.37},
 {"bus_id": 8, "vm_pu": 1.020, "va_deg": -13.97},
 {"bus_id": 9, "vm_pu": 1.010, "va_deg": -14.56},
 {"bus_id": 10, "vm_pu": 1.010, "va_deg": -15.12},
 {"bus_id": 11, "vm_pu": 1.010, "va_deg": -15.67},
 {"bus_id": 12, "vm_pu": 1.010, "va_deg": -16.21},
 {"bus_id": 13, "vm_pu": 1.010, "va_deg": -16.74},
 {"bus_id": 14, "vm_pu": 1.010, "va_deg": -17.26}
],
 " ...[1408 chars más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=4.36e-02; F_MAE=26.09; faithful=0.0 %; solved=False (voltage_error); flags=-; error=-; tokens=3440;
↪ cost_usd=0.0162; wall=8.9 s

```

## 5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- LLM-only: V\_MAE entre 1.38e-02 (llm\_only\_forced:cot/case30) y 4.54e-01 (llm\_only\_forced:structured/case14) p.u. sobre los items que devolvieron números.
- 0.6 % (1/160) de abstención en LLM-only (respuesta con `converged=false` o arrays vacíos).
- En LLM-only, 0 items no resueltos por `convergence_mismatch` y 139 por `voltage_error`.
- 8 items con `json_parse_failed`: la respuesta LLM-only no fue JSON parseable.
- Solved más alto: `llm_only_forced:cot` 1.2 % (1/80); más bajo: `llm_only_forced:structured` 0.0 % (0/80).
- Ningún item reclamó éxito sobre un fallo (`claimed_success_on_failure=0`).
- Trazabilidad de números (Faith.) media en LLM-only: 6.2 % (sin tools, todo número es no trazable salvo los copiados de la request).
- Costo total 2.0979 USD; el método más caro fue `llm_only_forced:cot` con 1.1272 USD (54 %).
- Solved por modelo: `openrouter:openai/gpt-4o` 1.2 % (1/80); `openrouter:openai/gpt-4o-mini` 0.0 % (0/80).

## 6. Archivos

Rutas absolutas bajo `results_probe_forced.report.rescored.json` (cuando existe) es el archivo leído para las métricas.

```

report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↳ /power-flow-agent/benchmarks/results_probe_forced/openai_gpt-4o-mini/report.json
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGr
↳ ids-Code-v2/power-flow-agent/benchmarks/results_probe_forced/openai_gpt-4o-mini/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGri
↳ ds-Code-v2/power-flow-agent/benchmarks/results_probe_forced/openai_gpt-4o-mini/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrid
↳ s-Code-v2/power-flow-agent/benchmarks/results_probe_forced/openai_gpt-4o-mini/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-Fo
↳ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_probe_forced/openai_gpt-4o-mini/scoreboard_per_case.json
traces (80 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-Sm
↳ artGrids-Code-v2/power-flow-agent/benchmarks/results_probe_forced/openai_gpt-4o-mini/traces
report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↳ /power-flow-agent/benchmarks/results_probe_forced/openai_gpt-4o/report.json
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGr
↳ ids-Code-v2/power-flow-agent/benchmarks/results_probe_forced/openai_gpt-4o/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGri
↳ ds-Code-v2/power-flow-agent/benchmarks/results_probe_forced/openai_gpt-4o/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrid
↳ s-Code-v2/power-flow-agent/benchmarks/results_probe_forced/openai_gpt-4o/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-Fo
↳ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_probe_forced/openai_gpt-4o/scoreboard_per_case.json
traces (80 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-Sm
↳ artGrids-Code-v2/power-flow-agent/benchmarks/results_probe_forced/openai_gpt-4o/traces
log: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/po
↳ wer-flow-agent/benchmarks/results_probe_forced/logs/openai_gpt-4o-mini.log
log: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/po
↳ wer-flow-agent/benchmarks/results_probe_forced/logs/openai_gpt-4o.log

```